A Pre-Registered Synthetic Benchmark for Open-Set Radio Signal Classification under Transmitter-Level Splits

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***Abstract*—Open-set recognition asks a classifier to label known classes and to reject inputs from classes it never saw. We present a small synthetic benchmark for open-set radio signal classification in which every nuisance factor is explicit: each simulated transmitter has its own IQ imbalance, amplifier compression, phase noise and DC offset, and data are split by transmitter. Hypotheses and pass criteria were fixed before the runs and then tested again on five fresh seeds. With four known modulations and two held-out ones (4PAM and 64QAM), five post-hoc scores were compared on the same trained network. The outcome depends on the unknown class. For 4PAM, Euclidean distance to class centroids in embedding space reaches an AUROC of 0.990 ± 0.010 over ten seeds, while scores computed from the logits are inverted (energy: 0.123 ± 0.038). For 64QAM, no score works: the AUROC lies between 0.335 and 0.569. Mahalanobis scoring is below the Euclidean score in all ten seeds. Training against auxiliary outliers raised rejection almost only for 4PAM and only when 8PAM was among the outliers. Four of eight initial hypotheses replicated, two negative results replicated, one effect did not replicate, one was inconsistent between the two seed sets, and one of our own interpretations was refuted by a follow-up experiment. The benchmark is a measurement instrument on synthetic data, not a new method.**

***Index Terms*—Open-set recognition, radio signal classification, modulation classification, hardware impairments, out-of-distribution detection, pre-registration, reproducibility.**

I. Introduction

DEEP classifiers for radio signals are usually evaluated on test classes that were all present in training. A deployed monitor does not have that guarantee: new emitters, waveforms and interference types appear, and the useful answer for them is "unknown", not the nearest known label. Open-set recognition (OSR) addresses this case [1], [2].

OSR results on radio data are hard to interpret for two reasons. Public datasets entangle channel, hardware and capture session, so it is unclear what a rejection score reacts to. And hardware impairments are themselves learnable signatures, which is the basis of RF fingerprinting [3]; a split by frame may therefore let a model use device identity.

This paper goes the other way from realism. It studies a deliberately small synthetic benchmark in which each factor is controlled and published. Each transmitter draws its impairment parameters once; data are split by transmitter; unknown classes take no part in training, model selection or threshold calibration; and the threshold is set on known validation data at a fixed acceptance target. Inside this instrument we ask how standard open-set scores behave when the unknown classes are structurally related to the known ones (64QAM to 16QAM, 4PAM to BPSK).

The second concern of the paper is the reliability of its own conclusions. Every hypothesis and its pass criterion was written down before the corresponding runs. The first eight hypotheses were then tested again on five fresh seeds, and three follow-up hypotheses addressed weaknesses that an internal review found in the first analysis. We report what replicated, what did not, and where our interpretation was wrong.

The contributions are:

1) an open benchmark for open-set radio signal classification with explicit per-transmitter impairments and transmitter-level roles for training, model selection, calibration and test;
2) a comparison of five post-hoc scores on the same trained network per seed, reported per unknown class, and of an Objectosphere-style loss [4] under three auxiliary-outlier pools with the training seed held fixed;
3) a pre-registered analysis with a replication on fresh seeds, including two replicated negative results, one effect that did not replicate, and one refuted interpretation.

Code, pre-registrations, per-seed results and the scripts that generate every number and figure of this paper accompany the submission.

II. Related Work

OSR was formalised for visual classifiers by Scheirer *et al.* [1] and extended to deep networks by OpenMax [5]; Geng *et al.* survey the field [2]. The scores compared here are standard: the maximum softmax probability (MSP) baseline [6], the max-logit score [7], the energy score [8], and class-conditional distances in feature space, of which the Mahalanobis detector [9] is the best-known form. The Objectosphere loss [4] trains against auxiliary known outliers. Detection quality is summarised by the area under the ROC curve [10].

Synthetic datasets for modulation classification were introduced by O'Shea and West [11]; O'Shea *et al.* [12] trained deep classifiers on simulated and over-the-air signals and studied the effect of carrier-frequency offset, symbol rate and multipath. Open-set formulations have been applied to transmitter authorisation [13] and to modulation recognition with signal-domain features [14]. RF fingerprinting shows that impairments such as IQ imbalance and DC offset identify a device [3]; this is why the benchmark splits by transmitter. We do not propose a method and claim no superiority of any score beyond the conditions measured.

III. System Model and Problem Formulation

A. Signal model

A frame is a complex baseband sequence of $N = 512$ samples at 8 samples per symbol. Symbols are drawn uniformly from the constellation of the transmitter's modulation and shaped by a unit-energy root-raised-cosine filter (roll-off 0.35, span 6 symbols); the frame is a window taken after the filter transient and normalised to unit mean power. The transmitter then applies, in this order,

$$y_1[n] = x[n]\left(1 + (|x[n]|/A_{\mathrm{sat}})^{2p}\right)^{-1/(2p)}, \tag{1}$$

$$y_2[n] = \mu\, y_1[n] + \nu\, y_1^*[n], \tag{2}$$

$$y_3[n] = y_2[n]\, e^{j\theta[n]} + d, \tag{3}$$

where (1) is the Rapp amplifier model [15] with the saturation amplitude A_{sat} relative to unit RMS input, (2) is IQ imbalance with $\mu = (1 + ge^{-j\phi})/2$ and $\nu = (1 - ge^{j\phi})/2$, $\theta[n]$ is a Gaussian random walk (phase noise) and d is a DC offset. The frame then receives a carrier-frequency offset $f \sim \mathcal{U}(-0.01, 0.01)$ cycles per sample with a random initial phase, and complex white Gaussian noise at a stated signal-to-noise ratio (SNR), defined on the impaired, frequency-shifted signal.

A transmitter fixes its modulation and impairment parameters once: $g \sim \mathcal{U}(0.9, 1.1)$, $\phi \sim \mathcal{U}(-5°, 5°)$, real and imaginary parts of d from $\mathcal{U}(-0.05, 0.05)$, phase-noise standard deviation log-uniform in $[2 \times 10^{-3}, 10^{-2}]$ rad per sample, $A_{\mathrm{sat}} \sim \mathcal{U}(1.0, 1.6)$ and $p \sim \mathcal{U}(2, 3)$. An *ideal* device sets all of these to their identity values. No fading, multipath or receiver impairment is simulated.

B. Classes and roles

The known classes are BPSK, QPSK, 8PSK and 16QAM. The unknown classes, used only at test time, are 64QAM and 4PAM. Two further classes, 16PSK and 8PAM, serve as auxiliary training outliers. All frames of a device stay in one role: training, validation for model selection, validation for calibration, or test. Unknown-class frames influence no fit, no model selection and no threshold.

C. Decision rule

A score $s(\mathbf{x})$, larger meaning more unknown-like, is compared with a threshold τ: the frame is accepted if $s(\mathbf{x}) \le \tau$ and then receives the label of the largest logit. τ is the smallest value that accepts at least 95% of the known calibration frames.

IV. Scores and Training Variants

The embedding network (CNN1D) has three blocks of convolution (kernel 7), batch normalisation, ReLU and max pooling with 32, 64 and 128 channels, followed by global average pooling and a linear layer; the embedding dimension is 128. With logits $\mathbf{z}$, embedding $\mathbf{e}$ and class centroids $\mathbf{m}_c$ computed from the known training frames, the post-hoc scores are

$$s_{\mathrm{MSP}} = 1 - \max_c \operatorname{softmax}(\mathbf{z})_c, \tag{4}$$

$$s_{\mathrm{logit}} = -\max_c z_c, \qquad s_{\mathrm{energy}} = -\log \textstyle\sum_c e^{z_c}, \tag{5}$$

$$s_{\mathrm{Euc}} = \min_c \|\mathbf{e} - \mathbf{m}_c\|_2, \tag{6}$$

$$s_{\mathrm{Mah}} = \min_c \left((\mathbf{e} - \mathbf{m}_c)^{\mathsf{T}} \boldsymbol{\Sigma}_\alpha^{-1} (\mathbf{e} - \mathbf{m}_c)\right)^{1/2}, \tag{7}$$

with $\boldsymbol{\Sigma}_\alpha = (1-\alpha)\boldsymbol{\Sigma} + \alpha\,(\operatorname{tr}\boldsymbol{\Sigma}/128)\,\mathbf{I}$, where $\boldsymbol{\Sigma}$ is the pooled within-class covariance of the known training embeddings and α is the shrinkage; $\alpha = 0.1$ unless stated. At $\alpha = 1$ the Mahalanobis score ranks frames exactly as the Euclidean score does.

The outlier-exposure variants add outlier frames to the training set and use a loss in the style of the Objectosphere loss [4]: known frames are pushed to a logit norm above a margin $\xi = 10$, and outlier frames toward uniform logits of small norm, with weight $\lambda = 0.1$. Our loss acts on the norm of the *logits* and the score is s_{MSP}; this differs from the cited method, which acts on the deep-feature magnitude and uses it in the score. We therefore write "Objectosphere-style". Three outlier pools are compared: both auxiliary classes (800 frames), 16PSK only and 8PAM only (400 frames each).

V. Experimental Setup

A. Data, training and metrics

Per seed and known class there are 40 training, 10 validation and 10 test devices, with 20, 10 and 10 frames per device; for the open-set task the validation devices are halved by device into model selection and calibration. There are 20 devices per auxiliary class and 10 per unknown class. This gives 3200 known training frames, 200 calibration frames, and 400 known and 200 unknown test frames per seed (100 per unknown class). SNRs are drawn from -4 to 20 dB in steps of 2 dB. Networks are trained with Adam (learning rate 10^{-3}, weight decay 10^{-4}, batch size 64) for at most 25 epochs, with early stopping (patience 5) on the accuracy over the model-selection devices.

The AUROC takes the unknown class as positive [10]; per unknown class, it compares that class with all known test frames. The unknown rejection rate is the fraction of unknown test frames with $s > \tau$; the known acceptance rate is the fraction of known test frames with $s \le \tau$. The open-set classification rate (OSCR) curve [4] plots the fraction of known test frames that are accepted and correctly labelled against the fraction of unknown frames accepted, as τ varies; we report its area. Frames of one device are correlated, so the seed is the unit of replication: all dispersions are sample standard deviations over seeds.

Two companion tasks validate the simulator. Closed-set modulation classification is run with impaired and with ideal devices, for the CNN1D and for a multilayer perceptron (MLP; hidden widths 256 and 128, dropout 0.1) on the raw frame; models are selected on all 10 validation devices per class and tested with 10 frames per test device at each SNR from -6 to 20 dB. A fingerprinting task asks a one-dimensional ResNet

[16] (base width 16) to identify 10 devices that all send QPSK; here the device is the label, so the split is by frame (600 test frames per seed), and an ideal-device control must stay at the chance level of 0.10.

B. Pre-registration and replication

The study has three campaigns. Campaign 1 trains each open-set variant separately and runs the companion tasks (H1–H4). Campaign 2 trains one cross-entropy network per seed, evaluates the five post-hoc scores on it, and trains the three outlier pools (H5–H8). Campaign 3 retrains the three pools with one shared training seed, so that only the pool differs (H7s); scores the campaign-1 classifiers on new frames of their training devices (H9, H10); and sweeps the shrinkage α, which is exploratory.

Campaigns 1 and 2 first ran on seeds 0–4 and were then repeated with unchanged code on fresh seeds 5–9. Each hypothesis predicts the sign of a per-seed paired difference. It is *supported* on a set of five seeds if the predicted sign occurs in at least four, and on ten seeds if it occurs in at least eight. A hypothesis is *replicated* if it is supported on the original seeds and the fresh seeds meet the four-of-five criterion on their own, and *not replicated* if it is supported on the original seeds only. Two further labels are ours and were not in the plans: *refuted*, when the opposite sign occurs in at least eight of ten seeds, and *inconsistent*, when the original seeds do not support the hypothesis, the fresh seeds do, and the pooled count is below eight. With five or ten seeds we report counts, not significance; for reference, the two-sided sign-test p-value is 0.002 at 10/10, 0.021 at 9/10, 0.109 at 8/10 and 0.344 at 7/10.

The analysis plans of campaigns 2 and 3 and of the replication were committed and pushed to the repository before their runs started. The record for campaign 1 is weaker. Its plan was written before the run, but committed about 4.5 minutes after the run began, when the first seed had finished, and pushed after the run ended; the file was not changed afterwards. In addition, one seed of each campaign-1 stage (seeds 2, 3 and 4) had been run as a single-seed pilot before the plan was written, as the plan states. The hypotheses of campaign 1 are among those retested on fresh seeds under a plan committed beforehand. All runs used PyTorch on four CPU threads; the five campaign runs took 1045, 749, 1093, 505 and 1176 s. No run failed and no seed was excluded.

VI. Results

Unless stated otherwise, values are means and sample standard deviations over seeds 0–9. Table I lists every hypothesis with its counts.

A. Post-hoc scores on one network (H1, H5, H6)

Table II and Fig. 1 give the results with both unknown classes pooled, as the hypotheses were stated. On the same network, the three scores computed from the logits have a mean AUROC below 0.5: 0.353 ± 0.088 for MSP, 0.278 ± 0.065 for max-logit and 0.286 ± 0.068 for energy. Pooled, max-logit and energy are below 0.5 in every seed (maxima 0.368 and 0.386); MSP is below 0.5 in nine seeds and reaches 0.509 in one. A value below 0.5 means that the unknown frames receive, on average, lower scores than the known ones, that is, more confident outputs. The Euclidean distance reaches 0.779 ± 0.020. H1, tested in campaign 1 with separately trained networks, is supported in 5/5 original and 5/5 fresh seeds (mean AUROC difference $+0.433$); campaign 2 shows the same gap on a single network, so it is not an effect of training variance.

TABLE I
Pre-registered hypotheses. Counts are seeds with the predicted sign. H7s, H9 and H10 were run on ten seeds at once. The outcome labels are defined in Section V.

	Prediction	0–4	5–9	All	Outcome
H1	Euclidean > MSP (AUROC)	5/5	5/5	10/10	replicated
H2	outlier training > MSP (rejection)	5/5	5/5	10/10	replicated
H3	fingerprint: impaired > ideal	5/5	5/5	10/10	replicated
H4	CNN1D: ideal > impaired	3/5	4/5	7/10	inconsistent
H5	Mahalanobis > Euclidean	0/5	0/5	0/10	refuted
H6	energy > MSP	2/5	0/5	2/10	refuted
H7	4PAM rej.: 8PAM > 16PSK pool	5/5	5/5	10/10	replicated
H7s	as H7, shared training seed	–	–	10/10	supported
H8	MSP AUROC: low > high SNR	4/5	3/5	7/10	not replicated
H9	MLP: seen > unseen devices	–	–	6/10	not supported
H10	CNN1D: seen > unseen devices	–	–	9/10	supported

The pooled values hide a split between the two unknown classes (Table III). For 4PAM the two embedding-space scores are almost perfect: the Euclidean distance has an AUROC of 0.990 ± 0.010 and rejects 0.938 ± 0.051 of the 4PAM frames, while the logit-based scores are inverted, most strongly max-logit and energy (0.127 ± 0.036 and 0.123 ± 0.038). For 64QAM no score works. The best mean AUROC is 0.569 ± 0.039 (Euclidean), with a rejection rate of 0.087 ± 0.051; the others lie between 0.335 and 0.513, and max-logit and energy reach 0.5 or more on 64QAM in 3/10 and 3/10 seeds. The pooled Euclidean AUROC is thus the average of a class that is detected and a class that is not. Fig. 2 shows the pooled score distributions for one seed and Fig. 3 the corresponding OSCR curves.

Two predictions failed, and both failures replicated. H5 predicted that the Mahalanobis score would beat the Euclidean score. It was lower in all ten seeds (0/10 in the predicted direction; mean difference -0.031). H6 predicted that the energy score would beat MSP. It did so in 2/5 original and 0/5 fresh seeds, and was lower in 8/10 (mean difference -0.067).

B. Auxiliary outliers (H2, H7, H7s)

Training against both auxiliary classes raises unknown rejection above the MSP baseline in every seed (H2, 10/10; 0.155 ± 0.057 against 0.028 ± 0.015 in campaign 1). The outlier class 8PAM is close to the unknown class 4PAM, which we recorded as a confound before the first run. Campaign 2 measured it by changing the outlier pool: rejection of 4PAM frames is 0.348 ± 0.225 with 8PAM outliers and 0.004 ± 0.005 with 16PSK outliers (H7, 5/5 original and 5/5 fresh seeds).

In campaign 2 the three pools were also trained with different seeds. Campaign 3 removed that difference. With one training seed for all pools, 4PAM rejection is 0.423 ± 0.189 with 8PAM outliers and 0.009 ± 0.014 with 16PSK outliers (H7s, 10/10; per-seed difference from $+0.050$ to $+0.670$). Rejection of

TABLE II

OPEN-SET RESULTS OVER TEN SEEDS, BOTH UNKNOWN CLASSES POOLED (MEAN ± STANDARD DEVIATION). THE FIVE POST-HOC SCORES USE THE SAME CROSS-ENTROPY NETWORK PER SEED; THE LAST THREE ROWS ARE SEPARATELY TRAINED NETWORKS SCORED WITH MSP. 400 KNOWN AND 200 UNKNOWN TEST FRAMES PER SEED; CALIBRATION TARGET 0.95.

Variant	AUROC	Unknown rejection	Known acceptance	OSCR area
MSP	0.353 ± 0.088	0.028 ± 0.015	0.951 ± 0.029	0.333 ± 0.083
Max-logit	0.278 ± 0.065	0.033 ± 0.019	0.944 ± 0.028	0.249 ± 0.055
Energy	0.286 ± 0.068	0.034 ± 0.014	0.943 ± 0.026	0.252 ± 0.057
Euclidean distance	0.779 ± 0.020	0.512 ± 0.034	0.954 ± 0.016	0.608 ± 0.060
Mahalanobis ($\alpha = 0.1$)	0.748 ± 0.024	0.472 ± 0.017	0.947 ± 0.022	0.587 ± 0.060
Outlier training, 16PSK + 8PAM	0.536 ± 0.061	0.194 ± 0.067	0.945 ± 0.024	0.464 ± 0.059
Outlier training, 16PSK	0.312 ± 0.070	0.033 ± 0.024	0.941 ± 0.025	0.289 ± 0.059
Outlier training, 8PAM	0.581 ± 0.068	0.206 ± 0.129	0.936 ± 0.033	0.487 ± 0.044

TABLE III

THE SAME RUNS AS TABLE II, PER UNKNOWN CLASS (MEAN ± STANDARD DEVIATION OVER TEN SEEDS; 100 FRAMES PER UNKNOWN CLASS AND SEED). THE AUROC COMPARES ONE UNKNOWN CLASS WITH ALL KNOWN TEST FRAMES.

	AUROC		Unknown rejection	
Variant	64QAM	4PAM	64QAM	4PAM
MSP	0.335 ± 0.076	0.370 ± 0.116	0.046 ± 0.028	0.009 ± 0.012
Max-logit	0.428 ± 0.103	0.127 ± 0.036	0.053 ± 0.030	0.013 ± 0.018
Energy	0.449 ± 0.111	0.123 ± 0.038	0.053 ± 0.021	0.015 ± 0.020
Euclidean distance	0.569 ± 0.039	0.990 ± 0.010	0.087 ± 0.051	0.938 ± 0.051
Mahalanobis ($\alpha = 0.1$)	0.513 ± 0.048	0.983 ± 0.006	0.035 ± 0.021	0.910 ± 0.030
Outlier training, 16PSK + 8PAM	0.408 ± 0.045	0.665 ± 0.112	0.058 ± 0.045	0.331 ± 0.117
Outlier training, 16PSK	0.387 ± 0.082	0.237 ± 0.087	0.062 ± 0.050	0.004 ± 0.005
Outlier training, 8PAM	0.436 ± 0.057	0.726 ± 0.124	0.064 ± 0.050	0.348 ± 0.225

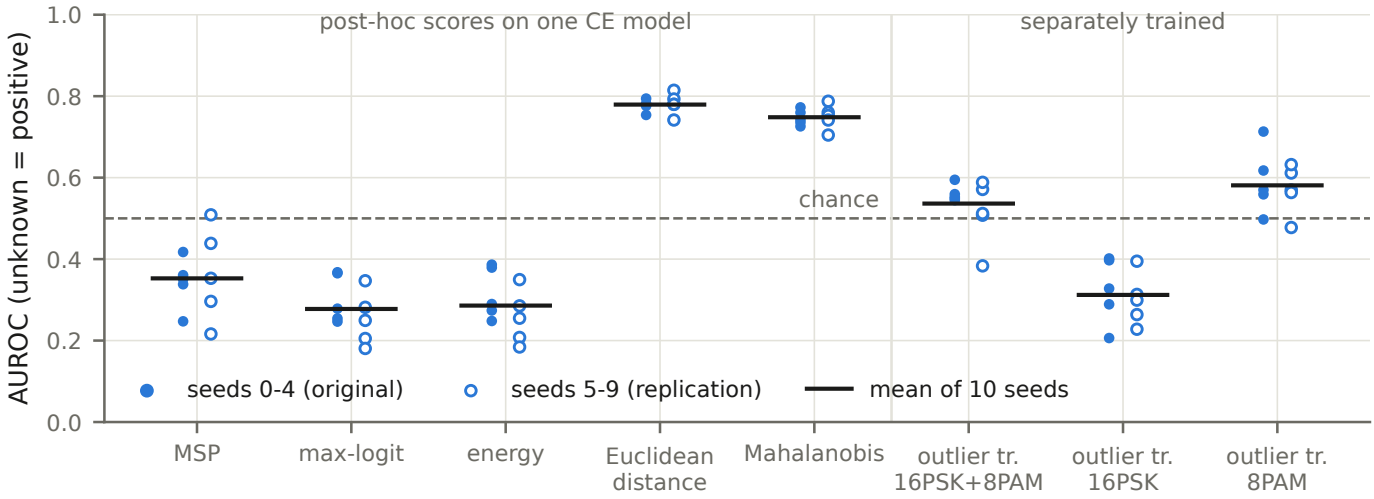


Fig. 1. AUROC of every open-set variant for each seed, both unknown classes pooled. Filled markers are the original seeds 0–4, open markers the replication seeds 5–9, and the bar is the mean of all ten.

64QAM frames stays low under every pool (0.058 ± 0.045, 0.037 ± 0.026 and 0.088 ± 0.054; Fig. 4). In this benchmark the gain from outlier training therefore appeared only with 8PAM in the pool and almost only on 4PAM. It is also unstable: with 8PAM outliers the per-seed 4PAM rejection ranges from 0.050 to 0.680. The pool with 16PSK alone gives a pooled AUROC of 0.273 ± 0.073. Because the design has one outlier pair and one unknown pair, it cannot separate similarity from other properties of 8PAM.

C. Dependence on SNR (H8)

H8 predicted that the MSP AUROC would be higher at low SNR (≤ 2 dB) than at high SNR (≥ 10 dB). It was supported on the original seeds (4/5) and *not replicated* on the fresh ones (3/5; pooled 7/10, below the criterion). Over ten seeds the two bands give 0.366 ± 0.082 and 0.338 ± 0.097. We conclude that the benchmark does not show an SNR dependence of the MSP result. Fig. 5 shows the AUROC per SNR; the cells are small (17 to 49 known and 3 to 27 unknown frames per SNR and seed), and the mean Euclidean AUROC is above the mean MSP AUROC at every SNR.

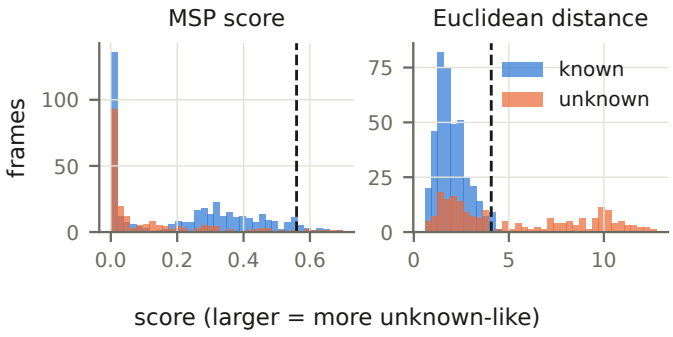


Fig. 2. Scores of the known and unknown test frames on one network (seed 0, 400 known and 200 unknown frames, both unknown classes). The dashed line is the calibrated threshold; frames to its left are accepted.

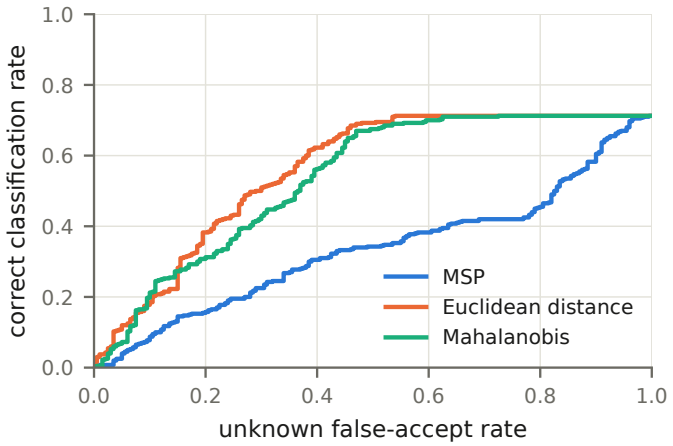


Fig. 3. OSCR curves of three scores on one network (seed 0).

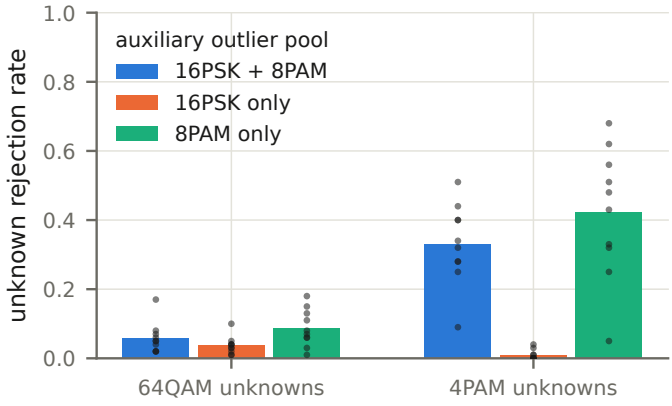


Fig. 4. Unknown rejection per unknown class for three outlier pools trained with one shared seed (campaign 3). Bars are means over ten seeds, dots are seeds; 100 frames per unknown class and seed.

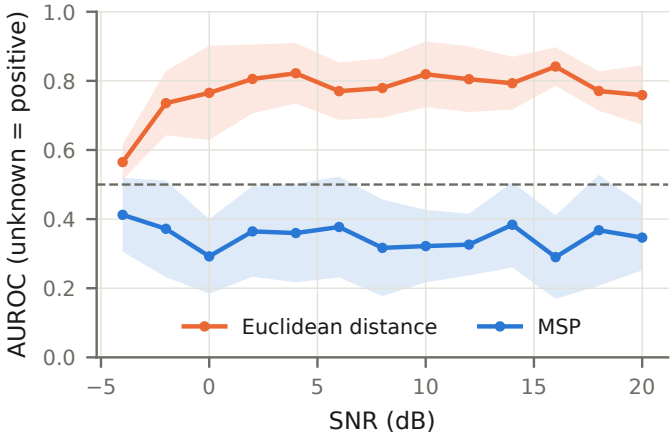


Fig. 5. AUROC per SNR for MSP and Euclidean distance on the same network, both unknown classes pooled; mean and standard deviation over ten seeds.

D. Validation of the simulator (H3, H4)

With impaired devices the fingerprinting network identifies the device with accuracy 0.221 ± 0.052; with ideal devices it scores 0.099 ± 0.011 (H3, 10/10). The second pre-registered clause of H3 also holds: the ideal-device accuracy lies within three binomial standard errors of the chance level 0.10 (0.063 to 0.137) in 10/10 seeds. The impairment chain therefore carries the device identity and nothing else in the pipeline does.

H4 predicted that removing the impairments would raise the closed-set accuracy of the CNN1D. The accuracies are 0.770 ± 0.019 for ideal and 0.749 ± 0.032 for impaired devices. The predicted sign occurred in 3/5 original and 4/5 fresh seeds, 7/10 in all, which is below the ten-seed criterion; the mean difference is $+0.022$. If the impairments cost closed-set accuracy at these magnitudes, the cost is about two points and not consistently detectable with ten seeds. Fig. 6 shows the accuracy against SNR.

E. What the MLP memorises (H9, H10)

The MLP reaches a training accuracy of 0.868 ± 0.129 but 0.251 ± 0.007 on unseen devices, the chance level of 0.25. An earlier draft of this paper read this as memorisation of the training *devices*, which a split by frame would hide. Campaign 3 tested that reading by scoring the same networks on new frames of the training devices (6720 frames per seed). H9 predicted a higher MLP accuracy there than on unseen devices. It was not supported (6/10; mean difference $+0.002$): the MLP scores 0.252 ± 0.006 on new frames of devices it was trained on. It memorises training frames, not devices, and fails on any new frame; the transmitter-level split is not what exposes it. Our earlier reading was wrong.

For the CNN1D the device shift is measurable but small: accuracy is 0.761 ± 0.033 on new frames of seen devices and 0.749 ± 0.032 on unseen devices (H10, 9/10; mean difference $+0.012$; Fig. 7).

F. Mahalanobis shrinkage (exploratory)

Because H5 was tested at one shrinkage value, campaign 3 recomputed the Mahalanobis score on the same networks for five values of α. This analysis was chosen after seeing campaign 2 and has no pass criterion. The pooled AUROC is 0.764, 0.748, 0.753 and 0.769 at $\alpha = 0.01$, 0.1, 0.5 and 0.9, against 0.779 for the Euclidean score, which $\alpha = 1$ reproduces exactly (Fig. 8). No tested amount of covariance whitening improved on the Euclidean score in the mean.

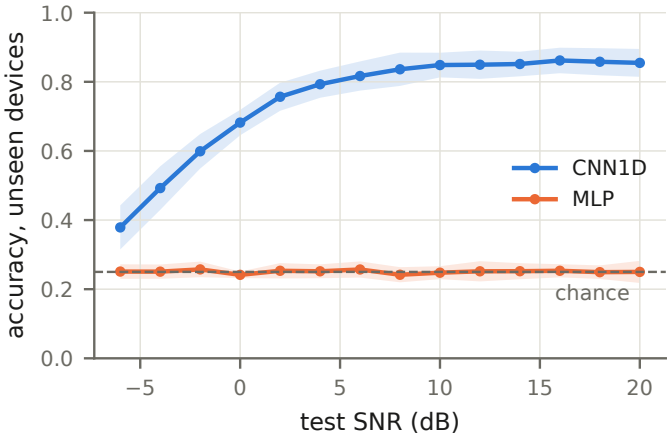


Fig. 6. Closed-set accuracy on unseen impaired devices against the test SNR; mean and standard deviation over ten seeds, 400 frames per SNR and seed.

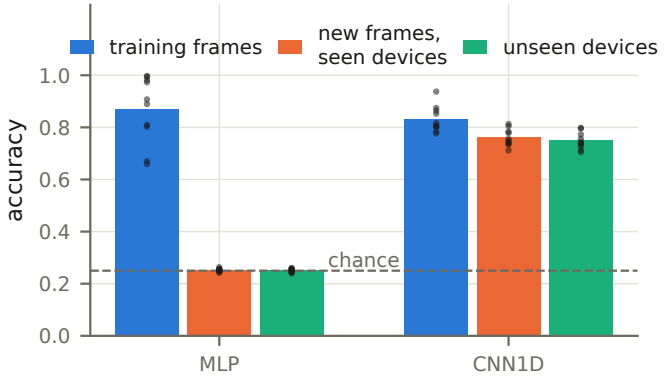


Fig. 7. Closed-set accuracy on the training frames, on new frames of the training devices and on unseen devices (campaign 3). Bars are means over ten seeds, dots are seeds.

VII. Discussion

The two unknown classes behave differently, and the pooled AUROC is not a good summary of either. 4PAM is detected almost perfectly by a distance in the embedding and is scored as *more* confident than known frames by every function of the logits; changing that function (H6) did not help, and max-logit and energy were more strongly inverted than MSP. 64QAM is not detected by any of the five scores. One reading is that 4PAM frames leave the region of the embedding occupied by the known classes while still producing confident logits, and that 64QAM frames, which contain the amplitude and phase structure of 16QAM, stay inside it. We did not test this reading: the predicted labels and embeddings of the unknown frames were not analysed, and it should be taken as a conjecture. What the measurements do show is that, on this network, the two distances in the embedding were the only scores that detected an unknown class, and that whitening by the within-class covariance (H5 and the sweep) did not improve on the plain Euclidean distance.

The outlier-pool experiment shows how much an outlier-exposure result can depend on the pool. Here the gain appeared only when 8PAM was in the pool and almost only on 4PAM. A result obtained with an outlier pool is hard to interpret without a statement of how the pool relates to the unknowns under test.

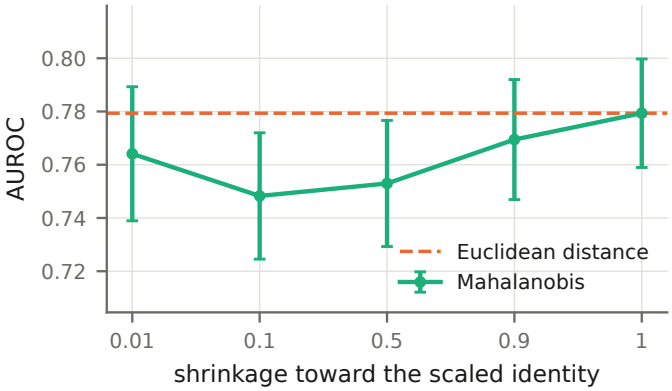


Fig. 8. Pooled Mahalanobis AUROC against the shrinkage α (exploratory); mean and standard deviation over ten seeds. The five tested values are equally spaced on the axis; $\alpha = 1$ equals the Euclidean score.

The first analysis was corrected in three places. Two came from the replication: H8, supported on the first five seeds with a small effect, did not survive fresh seeds, and the statement that MSP is below chance "in every seed" became nine of ten. The third came from the follow-up experiment, which refuted our reading of the MLP result. A fourth correction came from reporting per unknown class, as the plan of campaign 2 had listed: an earlier version of this text described the embedding as separating the unknown constellations in general, which holds for 4PAM only.

VIII. Limitations and Threats to Validity

Synthetic data. The transmitter model has no fading, multi-path, filter mismatch, timing error or amplifier memory, and no receiver model. Nothing here predicts performance on measured signals.

Two unknown classes. The conclusions rest on two unknown classes that behave in opposite ways. Which properties of an unknown class decide whether a distance in the embedding detects it cannot be answered with two classes.

One architecture and one configuration. One network, one SNR grid, one calibration target and one pair of loss parameters were used. The outlier-training loss acts on the logit norm and differs from the cited Objectosphere method.

Choice of classes. The known, unknown and auxiliary classes were fixed before the first run, but by us.

Statistical power. Ten seeds support only large effects. The criteria are sign counts, not significance tests, and effects of a few points (H4, H10) are at the edge of what the design resolves.

Pre-registration record. As stated in Section V, the plan of campaign 1 was committed after its run had started and pushed after it ended, and one seed per stage had been seen as a pilot. Its hypotheses were retested on fresh seeds under a plan committed beforehand. The labels "refuted" and "inconsistent" and the shrinkage sweep were not pre-registered.

Correlated frames. Frames of one device are correlated, so frame counts overstate the number of independent samples; per-SNR cells are small.

IX. Conclusion

In a synthetic benchmark with explicit transmitter impairments and transmitter-level splits, the result of open-set scoring depended on the unknown class. For 4PAM, the Euclidean distance to class centroids in the embedding reached an AUROC of 0.990 ± 0.010, while scores computed from the logits of the same network ranked 4PAM frames as more confident than known ones. For 64QAM, none of five scores exceeded a mean AUROC of 0.569. Mahalanobis scoring did not improve on the Euclidean distance at any tested shrinkage, and training against auxiliary outliers helped only with 8PAM in the pool and almost only on 4PAM. These findings held on fresh seeds. An SNR dependence of the MSP result did not, and our interpretation of the MLP's failure was refuted by a follow-up experiment.

The benchmark is small enough to rerun in under an hour and a half on four CPU threads, and its plans, per-seed results and generators are released so that it can be extended: with more unknown classes, with a channel model, with outlier pools that are disjoint from the unknowns by construction, with further architectures, and above all with a measured dataset that has per-device labels [3], [13].

Reproducibility

One command per campaign regenerates the per-seed results from the seeds; the numbers in this paper are generated from those results by a script, and the build fails if a number is missing. The repository also contains the three analysis plans with their commit history and an export of the data of one seed.

Disclosure

An AI assistant (Claude, Anthropic) was used for software development, experiment orchestration and drafting of this manuscript.

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